

ABNORMALITY REPORT — RCA & CAPA/PAA

Feed Charge Pump PU-2101B — Mechanical Seal Leakage

AR NUMBER AR-2026-ARP-0117	EQUIPMENT / TAG PU-2101B (Class B)	PLANT UNIT / DISCIPLINE ARP / Rotating (ROT)	OCCURRENCE DATE 12 Mar 2026
AR TYPE OPEDR (Operation) — Uptime	SEVERITY LEVEL Major	PRE-RISK ASSESSMENT III (Score 200)	RCA LEAD / PIC REL-05

DOWNTIME 18.5 h	PRODUCTION LOSS 251.6 ton	ESTIMATED LOSS \$226.44k
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PROBLEM STATEMENT Mechanical seal of Feed Charge Pump PU-2101B failed on 12-Mar-2026, causing seal leakage and an unplanned feed reduction to the ARP reactor section.	Immediate Action Isolate PU-2101B, swap to spare PU-2101A, replace mechanical seal cartridge
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Chronology of Events

Date / Time	Event	Evidence / Finding
11-Mar-2026	ARP produced resin grade PR-5502 (Reactor Rate ± 13.6 T/H).	Field operator noted minor seal weep on PU-2101B during routine round.
12-Mar-2026 02:15	Seal flush low-flow alarm annunciated;	visible feed seal leak with low suction pressure. Pump auto-tripped on low discharge pressure.
12-Mar-2026 02:40	Feed section reduced;	charge to reactor curtailed pending seal repair. Rotating team mobilized.
12-Mar-2026 09:20	Seal cartridge & shaft sleeve inspected — seal faces heat-checked / worn (dry-running signature).	
12-Mar-2026 20:45	New seal cartridge installed, pump commissioned, feed restored to ± 13.6 T/H.	

IMPACT SUMMARY (4W)

What: Mechanical seal of PU-2101B leaked / failed
When: 12-Mar-2026 (02:15)
Scope: Reduced charge feed to ARP reactor section
Uptime Loss: 18.5 hours → 251.6 ton → \$226.44k
Date Reported: 13 Mar 2026

Past Performance Analysis & Target Setting

HISTORICAL DATA & EVIDENCE

PU-2101B seal MTBF trended down from 26 to 9 months over the last 3 cycles. Seal-flush flow had been intermittently below the 6 L/min design minimum during feed swings, indicating recurring dry-run exposure.

TARGET SETTING

ACTUAL CONDITION (Before Improvement)

Feed reduction due to PU-2101B mechanical seal leak

TARGET CONDITION (Project Y)

No unplanned feed reduction due to PU-2101B seal failure (seal MTBF ≥ 24 months)

Recurrence of this equipment failure mode:
Before = 1 case → Target = 0 case

Parameter Verification (4P) & 4M + 1E Verification

#	Problem Phenomenon / Parameter	Result	Evidence / Finding
P1	Pump operated above rated flow	G	Operated at 13.1–13.7 T/H vs rated max 18 T/H — within limit.
P2	Suction pressure below NPSH margin	NG	Suction pressure dropped during feed swing; NPSH margin < 0.5 m (spec ≥ 1.0 m) — cavitation risk.
P3	Seal flush flow below minimum	NG	Seal-flush flow measured 3–4 L/min vs 6 L/min minimum — seal ran dry intermittently.
P4	Cooling water to seal cooler inadequate	G	Seal cooler outlet temp normal (42 °C) — cooling adequate.
P5	Seal faces beyond service life	G	Seal in service 11 months, within 24-month design life.
X1	No low-NPSH / flush-flow interlock in DCS	NG	No automatic trip or alarm on loss of seal-flush flow.
X2	Feed-swing SOP allows fast rate ramp	NG	SOP permits > 2 T/H per minute ramp, driving suction transients.
X3	PM covers seal only, not flush system	NG	Seal-flush plan (Plan 11) not on the PM check-sheet.
X4	Operator competency on seal-flush monitoring	G	Operators trained; no competency gap identified.

ROOT CAUSE

Premature mechanical seal failure caused by intermittent dry-running: suction cavitation during feed swings reduced NPSH and seal-flush flow dropped below the 6 L/min minimum, with no interlock to protect the seal.

NG = current condition not meeting standard (problem) • G = meets standard

Impact vs Control Prioritization

CONTROL / IMPACT	Low	Medium	High
Low			
Medium			X1 X2 X3
High			P2 P3

Prioritized root causes (high impact) are addressed first via Corrective & Pro-Active Actions →

Corrective & Pro-Active Action (CAPAA)

RC	Action	Corrective / Pro-Active / Preventive Action	Plan Date	PIC	Status
P3	Corrective	Install seal-flush flow switch with DCS alarm on PU-2101B	24-Apr-2026	REL-05	In Progress
X1	Corrective	Add low-NPSH / loss-of-flush trip interlock in DCS	24-May-2026	REL-02	Open
P3	Pro-Active	Roll out flush-flow switch to PU-2101A spare	24-May-2026	REL-05	Open
X1	Pro-Active	Apply interlock standard to all API-682 seal pumps in ARP	24-Jul-2026	REL-02	Open
P2	Preventive Suction cavitation	Revise feed-swing SOP to cap ramp at 1 T/H per minute	24-Apr-2026	ROT-01	
X3	Preventive Flush system not in PM	Add seal-flush system (Plan 11) to PU-2101B PM check-sheet	24-Apr-2026	REL-05	

RISK ANALYSIS OF CORRECTIVE ACTION

Install flow switch & interlock

Potential Risk: Nuisance trips during commissioning

Countermeasure: Stage commissioning with bypass timer & operator brief

24-May-2026 | REL-05

PM SCHEDULE ESTABLISHED

PM-1 Weekly Weekly seal-flush flow & seal-pot level check PU-2101B (Group: Rotating)

PM-2 3-monthly Quarterly seal condition & shaft run-out inspection (Group: Rotating)

Downtime & Closure Summary

KEY CORRECTIVE / PRO-ACTIVE ACTIONS

- Install seal-flush flow switch with DCS alarm on PU-2101B
- Add low-NPSH / loss-of-flush trip interlock in DCS
- Revise feed-swing SOP to cap ramp at 1 T/H per minute

DOWNTIME	PRODUCTION LOSS	ESTIMATED LOSS
18.5 h	251.6 ton	\$226.44k